

MULTIPLEXING & MULTIPLE ACCESS

Complete Study Notes with Mathematical Problems

FDM · TDM · TDMA · OFDM · CDMA

Contents

1. Multiplexing vs Multiple Access	2
2. Frequency Division Multiplexing (FDM)	2
3. Time Division Multiplexing (TDM)	4
4. Framing & Synchronization	5
5. TDMA — Time Division Multiple Access	7
6. OFDM — Orthogonal Frequency Division Mod	8
7. CDMA — Code Division Multiple Access	10
8. Worked Mathematical Problems	13
9. Comparison Summary	15

1. Multiplexing vs Multiple Access

1.1 Multiplexing

Definition: Multiplexing combines multiple analog or digital signals into a single composite signal for transmission over a *shared medium*. The transmitter combines all signals; the receiver separates them.

- Analogy: a multi-lane highway — all vehicles share one road.
- The sender (MUX) combines signals; the receiver (DEMUX) separates them.

1.2 Multiple Access

Definition: Multiple Access manages how *multiple independent users* share a single common transmission medium, coordinating who transmits when/where.

1.3 Key Differences

Aspect	Multiplexing	Multiple Access
Who controls?	One central system/device	Multiple independent users
Purpose	Combine signals	Share medium among users
Signal origin	Same location	Different locations
Examples	Cable TV, telephone trunk	WiFi, GSM mobile network

2. Frequency Division Multiplexing (FDM)

2.1 Core Concept

FDM divides the total available bandwidth into **non-overlapping frequency bands**. Each signal is modulated onto a different **subcarrier frequency** and occupies its own band simultaneously.

2.2 How FDM Works — Step by Step

Step 1 — Original Signals

Suppose three signals each occupy 0–5 kHz. Transmitting them directly would cause complete overlap and interference.

Signal 1: 0 – 5 kHz
Signal 2: 0 – 5 kHz
Signal 3: 0 – 5 kHz ← all identical bands → cannot co-exist!

Step 2 — Modulation with Different Carrier Frequencies

Signal 1 $\times \cos(2\pi \cdot 100 \text{ kHz} \cdot t)$ → spectrum shifted to 97.5 – 102.5 kHz
Signal 2 $\times \cos(2\pi \cdot 110 \text{ kHz} \cdot t)$ → spectrum shifted to 107.5 – 112.5 kHz
Signal 3 $\times \cos(2\pi \cdot 120 \text{ kHz} \cdot t)$ → spectrum shifted to 117.5 – 122.5 kHz

For AM: $s_k(t) = m_k(t) \cdot \cos(2\pi f_{ck} t)$ Bandwidth of each channel = $2 \times B_m$

Step 3 — Guard Bands

| -- Ch1 (97.5-102.5) -- | -- guard -- | -- Ch2 (107.5-112.5) -- | -- guard -- | -- Ch3 --
100 kHz 110 kHz 120 kHz

Guard bands are small unused frequency strips between adjacent channels. They prevent **inter-channel interference (ICI)** and ease bandpass filtering at the receiver.

Step 4 — Combine and Transmit

$$s_{\text{FDM}}(t) = s_1(t) + s_2(t) + s_3(t)$$

Step 5 — Receiver Separation

BPF centred at 100 kHz → demodulate → Signal 1
BPF centred at 110 kHz → demodulate → Signal 2
BPF centred at 120 kHz → demodulate → Signal 3

2.3 Total Bandwidth Calculation

$$\text{Total BW} = N \times (2 \cdot B_m) + (N-1) \times B_{\text{guard}}$$

Example: $N=3$ signals, $B_m=5$ kHz, $B_{\text{guard}}=2$ kHz

$$\text{Total BW} = 3 \times 10 \text{ kHz} + 2 \times 2 \text{ kHz} = 30 + 4 = 34 \text{ kHz}$$

2.4 Advantages and Disadvantages

Advantage	Disadvantage
Simple concept	Guard bands waste bandwidth
All signals simultaneous (no waiting)	Inter-channel interference possible
Ideal for analog signals	Fixed bandwidth — cannot adapt to traffic
No synchronisation needed	Reserved band unused when channel is idle

2.5 Real-World Applications

Application	How FDM is Used
FM Radio	Each station: separate carrier (e.g. 88.1, 92.4 MHz)
Cable TV	Each channel: separate frequency band
DSL Internet	Voice (0–4 kHz) and data (>25 kHz) in same copper pair
PSTN Trunk	Multiple voice calls multiplexed over a single cable

3. Time Division Multiplexing (TDM)

3.1 Core Concept

TDM divides the **time axis** into slots and assigns each channel its own slot in round-robin fashion. All channels share the same frequency; only one transmits at any instant.

If 1 second is shared among N channels equally:
Slot duration per channel = $1 / N$ seconds

Example: $N=2 \rightarrow$ Channel A gets 0–0.5 s, Channel B gets 0.5–1.0 s

3.2 Pulse Modulation Types

PAM — Pulse Amplitude Modulation

The amplitude of each pulse is proportional to the instantaneous value of the analog signal at the sampling instant. Different signals interleave their pulses in the gaps left by each other.

PAM sample: $p_k(nT_s) = m(nT_s)$
Sampling theorem: $f_s \geq 2 \cdot f_{\max}$ (Nyquist rate)

PWM — Pulse Width Modulation

The width (duration) of each pulse varies with the amplitude of the analog signal at the sample point. Larger amplitude $\rightarrow$ wider pulse.

3.3 Bit Interleaving vs Word Interleaving

(a) Bit Interleaving

Channels: A B C D
Output: A1 B1 C1 D1 | A2 B2 C2 D2 | A3 B3 C3 D3 | A4 B4 C4 D4
 $\leftarrow$ one bit from each channel per round $\rightarrow$

- Simplest method; rotary-switch model.
- Problem: all channels must have the same bit rate.
- Incompatible with mixed-speed media (optical fibre + coaxial).

(b) Word Interleaving

Channels: A B C D
Output: A1A2A3A4 | B1B2B3B4 | C1C2C3C4 | D1D2D3D4
 $\leftarrow$ one full word from each channel per round $\rightarrow$

- One complete word (8 bits, for example) taken from each channel.
- Supports different channel bit rates.

- More suitable for mixed traffic (video + audio).

3.4 Handling Unequal Data Rates

When channel A has data rate $3r$ and channels B, C, D each have rate r , channel A receives 3 slots for every 1 slot given to B, C, or D.

Rate: $A=3r$, $B=C=D=r$

Output: A1 B1 A2 C1 A3 D1 | A4 B2 A5 C2 A6 D2 | ...

A gets 3× as many slots as each low-rate channel

Efficiency: $\eta = (\text{Useful data bits}) / (\text{Total bits transmitted})$

For the above: $\eta = (3r + 3r) / (6 \text{ slots} \cdot r) = 1.0$ (100% — no guard needed in TDM)

4. Framing & Synchronisation in TDM

4.1 Why Framing Is Needed

Without framing, the receiver cannot determine which bit belongs to which channel. Environmental factors (temperature, clock drift) can cause the receiver's timing to slip, corrupting all channels simultaneously.

4.2 T1 Carrier Frame Structure (Example)

```
1 Frame = 193 bits
= 1 framing bit (F) + 24 channels × 8 bits/channel

|F| Ch1 (8b) | Ch2 (8b) | Ch3 (8b) | ... | Ch24 (8b) |
↑
Framing bit alternates: 1,0,1,0,1,0 ...

Frame rate: 8000 frames/second → Line rate = 8000 × 193 = 1.544 Mbit/s
```

4.3 Frame Bit Roles

Bit	Name	Function
M ₁ / M ₂	Multiframe bit	M ₁ =0 identifies the master frame; M ₂ =1 for others
F ₁ / F ₂	Frame sync bit	Alternating 0,1 pattern; receiver checks sync health
C bit	Stuffing control	C=1 → current data slot is stuffed (dummy); ignore it
[48]	Data payload	48 data bits per channel segment in each sub-frame

4.4 Added-Digit Framing — Worked Example

Setup: 3 channels (A, B, C), each 4 bits/frame; 1 framing bit per frame.

Channel A data: 1010
Channel B data: 0101
Channel C data: 1001

Frame 1 (F=1):

F	A1	B1	C1	A2	B2	C2	A3	B3	C3	A4	B4	C4
1	1	0	1	0	1	0	1	0	0	0	1	1

Transmitted bits: 1 101 010 100 011

Frame 2 (F=0):

Transmitted bits: 0 ...

Receiver sees F pattern: 1, 0, 1, 0 ... ✓ SYNCHRONISED

Decode Frame 1 (positions after F bit):

Pos 1,4,7,10 → Channel A: 1, 0, 1, 0 ✓

Pos 2,5,8,11 → Channel B: 0, 1, 0, 1 ✓

Pos 3,6,9,12 → Channel C: 1, 0, 0, 1 ✓

4.5 Bit/Byte Stuffing

When a channel's clock runs slightly slow, it produces fewer bits than the frame expects. The system inserts a **dummy (stuffed) bit** and signals this with C=1. The receiver discards that slot, maintaining synchronisation without data corruption.

Frame 48: [C=0] [A: 10110010] ← valid data

Frame 49: [C=0] [A: 01001101] ← valid data

Frame 50: [C=1] [A: 00000000] ← STUFFED – receiver ignores this slot

Frame 51: [C=0] [A: 10100110] ← valid data resumes

5. TDMA — Time Division Multiple Access

5.1 TDM vs TDMA

Aspect	TDM	TDMA
Control	Centralised (single MUX)	Distributed (each user requests slots)
Users	Fixed, known in advance	Variable; based on traffic
Idle slots	Wasted	Reallocated to active users
Example	Telephone trunk (T1/E1)	GSM mobile network

5.2 Statistical Multiplexing Advantage

In practice, not all incoming channels are active simultaneously — some transmit data, others are idle. TDMA exploits traffic statistics to accept *more users than physical slots*, improving resource utilisation.

**If each user is active $p = 10\%$ of the time and total capacity = C channels:
Expected simultaneous users = $N \times p$ (must $\leq C$)**

Example: $C=8$ slots, $p=0.1 \rightarrow$ can serve up to ~ 80 users with low blocking probability

5.3 GSM Example

1 GSM carrier (200 kHz) $\rightarrow$ 8 TDMA time slots
Frame duration: 4.615 ms $\rightarrow$ slot duration: 576.9 μ s

Slot 1: User A transmits
Slot 2: User B transmits
...
Slot 8: User H transmits
Slot 1: User A transmits again (next frame)

Idle user $\rightarrow$ slot reassigned to another user (statistical gain)

6. OFDM — Orthogonal Frequency Division Multiplexing

6.1 Motivation — The Bandwidth Problem

Conventional FDM wastes bandwidth with guard bands. If we want to send 16 symbols using two carriers simultaneously, naïve FDM would double the required bandwidth. OFDM solves this by using **mathematically orthogonal subcarriers** that can overlap without interfering.

6.2 Orthogonality — The Key Property

Two sinusoids at frequencies f_1 and f_2 are orthogonal if:

$$\int_0^T \cos(2\pi f_1 t) \cdot \cos(2\pi f_2 t) dt = 0$$

This holds when $f_2 - f_1 = k / T$ (k integer, T = symbol period)

Minimum subcarrier spacing: $\Delta f = 1 / T_s$

6.3 OFDM — Step by Step

Step 1 — Serial-to-Parallel Conversion

Input (serial): 1 0 1 1 0 0 1 0 1 1 0 1 0 0 1 1 (16 bits)

Parallel streams (N=4 subcarriers):

Subcarrier 1: 1 0 1 1

Subcarrier 2: 0 0 1 0

Subcarrier 3: 1 1 0 1

Subcarrier 4: 0 0 1 1

→ Each subcarrier sends 4 bits in the SAME time as 16 serial bits → 4× faster!

Step 2 — QAM Modulation per Subcarrier

QAM symbol on subcarrier k:

$$s_k(t) = A_k \cdot \cos(2\pi f_k t) - B_k \cdot \sin(2\pi f_k t)$$

A_k = in-phase component (real), B_k = quadrature component (imaginary)

Both cos and sin channels used → doubles spectral efficiency

Step 3 — IFFT Implementation

Instead of N separate oscillators (expensive!), the N QAM symbols are treated as frequency-domain coefficients and converted to a time-domain signal using **IFFT (Inverse Fast Fourier Transform)**.

$$x[n] = \text{IFFT}\{X[k]\} = (1/N) \sum X[k] \cdot e^{j2\pi kn/N} \quad n=0,1,\dots,N-1$$

Receiver recovers symbols: $X[k] = \text{FFT}\{x[n]\}$

6.4 Cyclic Prefix (Guard Interval)

To prevent **Inter-Symbol Interference (ISI)** caused by multipath propagation, the last G samples of each OFDM symbol are copied and prepended as a cyclic prefix.

CP length $G \geq \text{maximum multipath delay spread } \tau_{\text{max}} \times f_s$

Overhead: $\eta = T_s / (T_s + T_{CP})$
Example: $T_s=64$ samples, $T_{CP}=16$ samples $\rightarrow \eta=64/80=80\%$

6.5 OFDM Summary Table

Parameter	Detail
Applications	5G NR, 4G LTE, Wi-Fi 802.11a/g/n/ac/ax, ADSL/VDSL, DVB-T
Subcarriers	5G uses up to 4096; Wi-Fi 802.11ax uses 2×996
Subcarrier spacing	15 kHz (4G LTE), 30/60/120/240 kHz (5G NR)
Guard interval	Cyclic prefix; typically 1/4 of symbol duration
Speed gain	Proportional to number of subcarriers vs. single-carrier

7. CDMA — Code Division Multiple Access

7.1 Core Concept

All users transmit simultaneously on the **same frequency band**, distinguished by unique **orthogonal spreading codes**. The receiver uses its code to extract only the desired signal; all others appear as noise.

7.2 Orthogonal Codes

Codes C_i , C_j , C_k are orthogonal if: $C_i \cdot C_j = 0$ ($i \neq j$)

Dot product: $C_i \cdot C_j = \sum c_{i,k} \cdot c_{j,k}$ over all chips

Self-correlation: $C_i \cdot C_i = N$ (where N = code length)

7.3 Voltage Convention

Binary 0 $\rightarrow$ +1 V | Binary 1 $\rightarrow$ -1 V

7.4 Encoding — Worked Example

User 1: data = 00, code = 0101
User 2: data = 10, code = 0011
User 3: data = 11, code = 0000

User 1 Encoding:

data = 00 $\rightarrow$ voltages: +1 +1 (bit 0 $\rightarrow$ +1)

Each data bit is spread over the 4-chip code:

Bit 0 (+1) $\rightarrow$ +1 repeated 4 times: +1 +1 +1 +1

But XOR with code 0101 ($\rightarrow$ +1 -1 +1 -1):

(+1)(+1) (+1)(-1) (+1)(+1) (+1)(-1) = +1 -1 +1 -1

Bit 0 again (second data bit): +1 -1 +1 -1

User 1 transmitted chips: +1 -1 +1 -1 +1 -1 +1 -1

User 2 Encoding:

```
data = 10 → bit 1 = -1V, bit 0 = +1V
code = 0011 → +1 +1 -1 -1
```

Bit 1 (-1) $\times$ code (+1 +1 -1 -1) = -1 -1 +1 +1
 Bit 0 (+1) $\times$ code (+1 +1 -1 -1) = +1 +1 -1 -1

User 2 transmitted chips: -1 -1 +1 +1 +1 +1 -1 -1

User 3 Encoding:

data = 11 $\rightarrow$ both bits = -1V
code = 0000 $\rightarrow$ +1 +1 +1 +1

Bit 1 (-1) $\times$ code (+1 +1 +1 +1) = -1 -1 -1 -1
 Bit 1 (-1) $\times$ code (+1 +1 +1 +1) = -1 -1 -1 -1

User 3 transmitted chips: -1 -1 -1 -1 -1 -1 -1 -1

7.5 Channel Signal (Superposition)

[illegible]

7.6 Decoding — Receiver 1 (code = 0101 → +1,-1,+1,-1)

[illegible]

Mathematical proof:

$$(m_{\text{CC}} + m_{\text{C}} + m_{\text{C}}) \cdot \text{C} = m_{\text{C}}(\text{C} \cdot \text{C}) + m_{\text{C}}(\text{C} \cdot \text{C}) + m_{\text{C}}(\text{C} \cdot \text{C})$$

$$= m_{\text{C}} \times N + m_{\text{C}} \times 0 + m_{\text{C}} \times 0 = m_{\text{C}} \times N$$

Divide by $N \rightarrow$ recover m_i exactly. Other users cancel due to orthogonality.

7.7 Decoding — Receiver 2 (code = 0011 → +1,+1,-1,-1)

Received: -1 -3 +1 -1 | +1 -1 -1 -3

Code C■: +1 +1 -1 -1 | +1 +1 -1 -1

[illegible]

Product: -1 -3 -1 +1 | +1 -1 +1 +3

Sum (first 4): $-1-3-1+1 = -4 \rightarrow -4/4 = -1V \rightarrow \text{bit } 1$

Sum (second 4): $+1-1+1+3 = +4 \rightarrow +4/4 = +1V \rightarrow \text{bit } 0$

Decoded data = 10 ✓ (User 2 sent 10)

7.8 Spread Spectrum & Security

Spread Spectrum: the encoded signal occupies a much wider bandwidth than the original data. An eavesdropper without the spreading code sees only broadband noise. CDMA is resistant to narrowband jamming — hence its importance in military communications.

8. Worked Mathematical Problems

Problem 1 — FDM Bandwidth Calculation

Problem: 5 voice signals, each 300–3400 Hz, are frequency-division multiplexed with 600 Hz guard bands between adjacent channels. (a) Find the bandwidth of each modulated channel. (b) Find the total transmission bandwidth.

Solution:

(a) Bandwidth of one voice signal = $3400 - 300 = 3100$ Hz
DSB-SC modulation doubles the spectrum around the carrier:
Channel BW = $2 \times 3100 = 6200$ Hz (or 6.2 kHz)

(b) $N = 5$ channels, guard band = 600 Hz each ($N-1 = 4$ gaps)
Total BW = $N \times \text{channel_BW} + (N-1) \times \text{guard_BW}$
 $= 5 \times 6200 + 4 \times 600$
 $= 31000 + 2400$
 $= 33\,400$ Hz = 33.4 kHz

Problem 2 — T1 Line Rate Verification

Problem: A T1 carrier multiplexes 24 voice channels, each sampled at 8000 samples/s with 8 bits/sample, plus 1 framing bit per frame. Calculate the T1 line bit rate.

Solution:

Samples per second per channel = 8000
Bits per sample = 8
Data bits per frame = $24 \text{ channels} \times 8 \text{ bits} = 192 \text{ bits}$
Framing bits per frame = 1
Total bits per frame = $192 + 1 = 193 \text{ bits}$

Frame rate = 8000 frames/second (Nyquist for 4 kHz voice)

Line rate = $193 \times 8000 = 1\,544\,000 \text{ bit/s} = 1.544 \text{ Mbit/s}$ ✓

Problem 3 — TDM Slot & Frame Duration

Problem: A TDM system carries 4 channels at 64 kbit/s each. The frame contains 1 sync bit plus one byte (8 bits) per channel. (a) Find the total line rate. (b) Find the frame period. (c) Find the slot duration.

(a) Bits per frame = $1 + 4 \times 8 = 33$ bits
 Each channel at 64 kbit/s, sampled $64\,000/8 = 8000$ frames/s
 Line rate = $33 \text{ bits} \times 8000 \text{ frames/s} = 264 \text{ kbit/s}$

(b) Frame period $T_f = 1 / 8000 = 125 \mu\text{s}$

(c) Slot duration = $T_f / 4 \text{ channels} = 125 / 4 = 31.25 \mu\text{s}$
 (including the sync bit amortised)

Problem: An OFDM system uses $N=64$ subcarriers, symbol duration $T_s=4 \mu s$, cyclic prefix $T_{CP}=0.8 \mu s$, and each subcarrier uses 16-QAM. (a) Find subcarrier spacing. (b) Find the useful bit rate. (c) Find efficiency.

Solution:

(a) Subcarrier spacing $\Delta f = 1/T_s = 1/4\mu s = 250 \text{ kHz}$
Total RF bandwidth = $N \times \Delta f = 64 \times 250 \text{ kHz} = 16 \text{ MHz}$

(b) 16-QAM $\rightarrow \log_2(16) = 4 \text{ bits/symbol/subcarrier}$
Symbols per second per subcarrier = $1 / (T_s + T_{CP})$
 $= 1 / (4 + 0.8)\mu s$
 $= 1 / 4.8\mu s$
 $\approx 208\,333 \text{ symbols/s}$
Total bit rate = $N \times 4 \times 208\,333$
 $= 64 \times 4 \times 208\,333$
 $\approx 53.33 \text{ Mbit/s}$

(c) Spectral efficiency = Bit rate / Bandwidth
 $= 53.33 \text{ Mbit/s} / 16 \text{ MHz}$
 $\approx 3.33 \text{ bit/s/Hz}$
Overhead factor = $T_s / (T_s + T_{CP}) = 4/4.8 \approx 0.833 \text{ (83.3\%)}$

9. Comprehensive Comparison Summary

Aspect	FDM	TDM	CDMA	OFDM
Resource divided	Frequency	Time	Code	Freq (orthogonal)
Simultaneous Tx?	Yes	No (round-robin)	Yes	Yes
Same frequency?	No	Yes	Yes	Yes (overlapping)
Guard needed?	Guard bands	Sync/framing bits	None	Cyclic prefix
Signal type	Analog ideal	Digital ideal	Digital	Digital
Sync required?	No	Yes (critical)	Yes (chip sync)	Yes (FFT timing)
Main use	Cable TV, radio	Telephony (T1/E1)	CDMA2000, military	4G/5G, Wi-Fi
Bandwidth efficiency	Low (guard waste)	High	Medium	Very high
Interference robust?	Moderate	Moderate	High (spread)	High (OFDM)

Key Formulas Reference

FDM Total BW: $BW_{total} = N \cdot BW_{ch} + (N-1) \cdot BW_{guard}$

TDM Line Rate: $R_{line} = (N \cdot b + 1) \times f_s \text{ bits/s}$
(N channels, b bits/channel, 1 framing bit, f_s frames/s)

CDMA Decode: $m_{data} = (1/N) \cdot (\text{received chips}) \cdot C_{code}$

OFDM Subcarrier Spacing: $\Delta f = 1/T_s$

OFDM Spectral Efficiency: $\eta_{SE} = [\log_2(M) \times N] / [(T_s + T_{CP}) \times N \times \Delta f] = \log_2(M) \times T_s / (T_s + T_{CP})$
bit/s/Hz

Memory Aids

FDM — Multi-lane highway: each car (signal) stays in its own lane (frequency band).

TDM — Traffic light junction: one car (channel) moves at a time; others wait their turn.

CDMA — A party where each pair speaks a different language; only the matching listener can understand their speaker; all others hear noise.

OFDM — A wide multi-lane highway where lanes slightly overlap, but all cars are perfectly steered (orthogonal) so collisions never occur.